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MH1200 Linear Algebra I

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1. Checking if a vector $\vec{u}$ is in the span of a set of vectors S

This reduces to checking if the following equation is satisfiable for some choice of coefficients c_i:

$$\sum_{i=1}^{n} c_i \vec{v_i} = \vec{u}$$

Which reduces to checking if the following augmented matrix is consistent:

$$\left(\begin{array}{cccccc|c} \uparrow & \uparrow & \uparrow & & \uparrow & \uparrow \\ \vec{v_1} & \vec{v_2} & \vec{v_3} & \cdots & \vec{v_n} & \vec{u} \\ \downarrow & \downarrow & \downarrow & & \downarrow & \downarrow \end{array}\right)$$

$\vec{u}$ is in the span of S iff the matrix above is satisfiable (i.e., consistent).

2. Finding a basis that spans the set of solutions of a homogenous system

Steps:
1. Write the system as an augmented matrix.
2. Reduce into REF or RREF.
3. Find a parameterization.
4. Separate the variables in a parameterization.

3. Comparing the spans of two lists of vectors

Suppose you have two sets of vectors S and T, with $\vec{v_i} \in S$ and $\vec{u_i} \in T$.

Then, $\text{span}(S) \supseteq \text{span}(T)$ iff reducing the following matrix to RREF does not result in a row where everything to the left of the bar is zero and there is a number to the right of the bar that is nonzero.

$$\left(\begin{array}{cccccc|cccc} \uparrow & \uparrow & \uparrow & & \uparrow & \uparrow & \uparrow & \uparrow & & \uparrow \\ \vec{v_1} & \vec{v_2} & \vec{v_3} & \cdots & \vec{v_n} & \vec{u_1} & \vec{u_2} & \vec{u_3} & \cdots & \overrightarrow{u_n} \\ \downarrow & \downarrow & \downarrow & & \downarrow & \downarrow & \downarrow & \downarrow & & \downarrow \end{array}\right)$$

Theorem Spanning Sets

Theorem 1

Suppose S, S' are two sets of spanning vectors, with $V = \text{span}(S)$ and $V' = \text{span}(S')$.

If $S \subseteq V'$, then $V \subseteq V'$.

4. Notes to Self:

- For your revision (and for ICPC training ig): Derive an algorithm to find the intersection of the spans of two sets of spanning vectors.